Topic: Line Integrals

I. Definition & Properties

Defn: A **Line Integral** is integration over a curve, C is given by

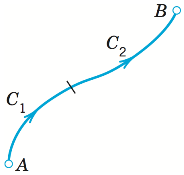
Vector-valued functions: $$\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = \underbrace{\int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt}_{\textit{computing form}} \qquad t \in [a, b]$$

Scalar-valued functions: $$\int_C g(\mathbf{r})\, ds = \overbrace{\int_a^b g(\mathbf{r}(t))\, |\mathbf{r}'(t)|\, dt} \qquad t \in [a, b]$$

Note: The expressions above are valid whether C is a planar or space curve.

Properties: Let $\mathbf{F}$ be vector-valued function and k a constant.

- $\displaystyle\int_C k\,\mathbf{F} \cdot d\mathbf{r} = k \int_C \mathbf{F} \cdot d\mathbf{r}$
- $\displaystyle\int_C (\mathbf{F} + \mathbf{G}) \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot d\mathbf{r} + \int_C \mathbf{G} \cdot d\mathbf{r}$
- $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r} \quad (C = C_1 \cup C_2)$



Ex: If $\mathbf{F} = [y^2, -x^2]$ and $C: y = 4x^2$ from $(0,0)$ to $(1,4)$, then compute $\int_C \mathbf{F} \cdot d\mathbf{r}$

Soln: We first need to parametrize C: Let $x = t \implies y = 4t^2$ so

$\mathbf{r}(t) = [t, 4t^2]$ and since $x = t$ and the path is from $(0,0)$ to $(1,4)$ then, $0 \le t \le 1$

$\mathbf{r}'(t) = [1, 8t]$ and $\mathbf{F}(\mathbf{r}(t)) = [(4t^2)^2, -(t)^2] = [16t^4, -t^2]$ Therefore

$$\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt = \int_0^1 [16t^4, -t^2] \cdot [1, 8t]\, dt = \int_0^1 \left(16t^4 - 8t^3\right) dt$$

$$= \left(\frac{16}{5} t^5 - \frac{8}{4} t^4\right)\Big|_0^1$$

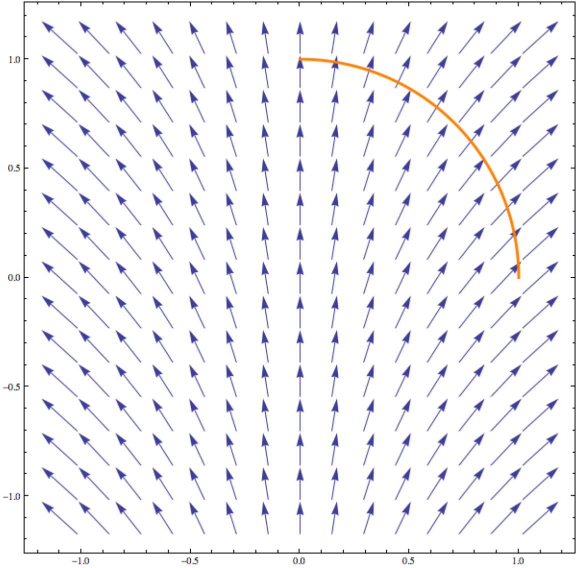
$$= \left(\frac{16}{5}(1)^5 - \frac{8}{4}(1)^4\right) - (0) = \frac{6}{5}$$

Ex: Determine $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = [x, y - z^2, z - x]$ and $C : \mathbf{r}(t) = [2\cos t, t, 2\sin t]$ from $(2, 0, 0)$ to $(2, 2\pi, 0)$

Soln:

II. Work

Suppose there was a mass moving through a planar force field, $\mathbf{F}$, along a path, C.



Force field:

$$\mathbf{F}(x, y) = [x, 1]$$

Parameterization of C:

$$\mathbf{r}(t) = [\cos t, \sin t], \quad 0 \leq t \leq \frac{\pi}{2}$$

What is the work done in moving this mass along C?

Previously, we calculated work using the dot product:

$$W = \mathbf{F} \cdot \mathbf{r}$$

However, since both the force and displacement change at every point along C, we are forced to add up "many" of "small works" at different points(i.e. integrate).

"Small" Work = (Force) ("Small" Displacement) = (Force) (Rate) ("Small" time)

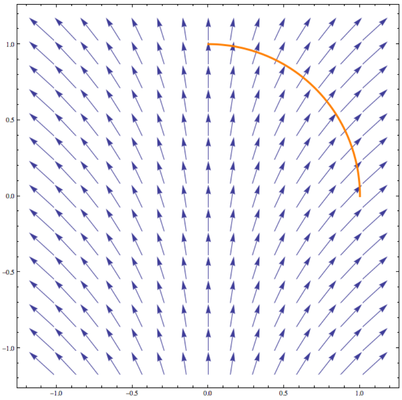
$$dW = \mathbf{F} \cdot d\mathbf{r} = \mathbf{F} \cdot \mathbf{r}'(t)\, dt$$

Total Work = Sum all the "Small Work" along C (Line Integral)

$$W = \int_C dW = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

Ex: $\mathbf{F} = [x, 1]$ with $\mathbf{r}(t) = [\cos t, \sin t]$, $0 \leq t \leq \frac{\pi}{2}$

What is the work done to move the object along this path when exposed to **F**?



$\mathbf{F}(\mathbf{r}(t)) = [\cos t, 1]$

$\mathbf{r}'(t) = [-\sin t, \cos t]$

$\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = -\sin t \cos t + \cos t$

$$\int_0^{\pi/2} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt = \int_0^{\pi/2} -\sin t \cos t + \cos t\, dt$$

$$= \int_0^{\pi/2} -\sin t \cos t\, dt + \int_0^{\pi/2} \cos t\, dt$$

$$= \int_{t=0}^{t=\pi/2} u\, du + \sin t\Big|_0^{\pi/2}$$

$$= \frac{u^2}{2}\Bigg|_{t=0}^{t=\pi/2} + 1$$

$$= \frac{\cos^2 t}{2}\Bigg|_0^{\pi/2} + 1 = -\frac{1}{2} + 1 = \frac{1}{2}$$

$u = \cos t$

$du = -\sin t\, dt$

Ex: Let $\mathbf{F} = [xy, 2y^2]$ represent a force vector field in the x-y plane. If a particle moves along a quarter circle from $(2,0)$ to $(0,2)$, then find the work done on that particle.

Want: $\displaystyle\int_C \mathbf{F}\cdot d\mathbf{r}$, Need: A parameterization for C

Since C is part of a circle, we should think of polar coordinates:

$$x = 2\cos t$$
$$y = 2\sin t \quad \Longrightarrow \quad \mathbf{r}(t) = [2\cos t, 2\sin t], \quad t \in \left[0, \frac{\pi}{2}\right]$$

$\mathbf{F}(\mathbf{r}(t)) = [4\sin t\cos t, 8\sin^2 t]$

$\mathbf{r}'(t) = [-2\sin t, 2\cos t]$

$\mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t) = -8\sin^2 t\cos t + 16\sin^2 t\cos t = 8\sin^2 t\cos t$

$$\int_a^b \mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)\,dt = \int_0^{\pi/2} 8\sin^2 t\cos t\,dt \qquad u = \sin t \Longrightarrow du = \cos t dt$$

$$= 8\int_{t=0}^{t=\pi/2} u^2\,du$$

$$= 8\,\frac{u^3}{3}\bigg|_{t=0}^{t=\pi/2} = \frac{8}{3}\sin^3 t\bigg|_0^{\pi/2} = \frac{8}{3}\left[\sin^3\left(\frac{\pi}{2}\right) - \sin^3(0)\right] = \frac{8}{3}$$

Ex: Suppose a wire has density $\delta(x, y, z) = 3\sqrt{5 + x}$. Thinking of the wire as a curve in space, it could be parameterized by $\mathbf{r}(t) = t\mathbf{i} + 2t\mathbf{j} + \frac{2}{3}t^{3/2}\mathbf{k}$ where $0 \leq t \leq 2$. Find the mass of the wire.

Note 1: Since the wire is assumed to be a curve, δ is a linear density $\left(\dfrac{Mass}{Length}\right)$

Note 2: Since δ is not constant, we must integrate where

$$small\,mass \longrightarrow dM = \delta\, ds \longleftarrow \left(\frac{Mass}{Length}\right)(small\,length)$$

$$\implies \quad Mass = \int_C \delta\, ds = \int_0^2 \delta(\mathbf{r}(t))\,|\mathbf{r}'(t)|\,dt$$

$\delta(\mathbf{r}(t)) = 3\sqrt{5 + t}$

$\mathbf{r}'(t) = [1, 2, t^{1/2}] \qquad \implies \qquad |\mathbf{r}'(t)| = \sqrt{1^2 + 2^2 + (t^{1/2})^2} = \sqrt{5 + t}$

$$Mass = \int_0^2 \delta(\mathbf{r}(t))\,|\mathbf{r}'(t)|\,dt = \int_0^2 \left(3\sqrt{5+t}\right)\left(\sqrt{5+t}\right)dt$$

$$= \int_0^2 3\,(5 + t)\; dt$$

$$= \left.\left(15t + \frac{3}{2}t^2\right)\right|_0^2 = 30 + \frac{12}{2} = 36$$